Software Testing and Design Patterns: Advanced Software Engineering
Assessment Guide

This document is meant to be used only as a guide. The Assessment Rubric at the end of your assessment brief is the only thing you will be assessed against so ensure that you are happy that you have met the requirements outlined there before you submit your assessment.

Hello and welcome to your assessment guide for Module 7: Software Testing and Design Patterns: Advanced Software Engineering . In this guide we will take a look at the assessment requirements and dive into what you are expected to do in order to achieve different grade levels. Let me first define some common terms that you will hear your instructor talking about and that you will see in this guide:

The grade ladder

When it comes to the grade that you are aiming for in your assignments I would like you to think about the assessment rubric as a ladder to climb (for grade D and up). The D grade criteria is the minimum needed to pass - this should be the minimum that you are required to include in your assessment. Each successive rung of this ladder is a new grade. Therefore the bottom rung for a pass is a D, to get a C you must include everything from a D plus what is required for a C. To get a B you must include everything that is required for a D and a C plus what is required for a B and so on. If you get a '+' grade (such as a C+) it means that you've nearly reached the next rung in the ladder but have just fallen short in evidencing everything. In this guide we will take a look at climbing this ladder for each of the Software Testing and Design Patterns: Advanced Software Engineering learning outcomes.

A note on *evidence*

In what follows you will see that you are asked to provide evidence for various things. Think carefully about how you might best provide this evidence to the assessor so that you can achieve the grade that you're aiming for. You are encouraged to make use of diagrams and visualisations where appropriate (for example when providing evidence that you have designed something). When it comes to code please include these as screen grabs but do not use an egregious amount of text in these screen grabs as this should be reserved for the main body of the document. Sometimes it can also be useful to organise information into tables, bullet points, or to use other formatting techniques to aid in the clarity of your writing. The main body of the report is where most of the information should lie and the additional pieces of evidence (diagrams, visualisations, screen grabs of code) are supplementary. This means you must use the main text to refer to them and to describe what is in them.

Climbing the ladder

Let us take a look at what you should do to climb the ladder for each of the learning outcomes in this assessment.

LO1 - Analyse the software development lifecycle to identify where quality is built in or absent, establishing a data-backed case for improvement

For this learning outcome, ...

F - Failed for now

- You don't analyse current practice or you don't locate it within any recognisable SDLC framework.

E - Almost at the passing grade

- You identify basic SDLC stages, but you describe practice without assessing quality or testing activity.

D - Congratulations, you passed!

- You describe current development practice and locate it within the SDLC.
- You outline testing activity, but you don't analyse quality implications or connect gaps to business impact.
- To strengthen this, explain the relationship between design practices, testing activity, and software quality outcomes, and identify specific quality gaps with reference to business context to establish a clear case for improvement.

C - Now we're climbing the ladder!

- You explain the relationship between current design practices, testing activity, and software quality outcomes.
- You identify specific quality gaps with reference to business context and establish a clear case for the improvements explored in the project.
- To move this on, analyse quality implications in depth using evidence from the organisation, connect gaps explicitly to specific SDLC stages, and show how design and testing decisions interact to produce quality outcomes.

B - Nearly at the top!

- You analyse the quality implications of current practice in depth, justifying gaps with evidence from the organisation and connecting them explicitly to specific SDLC stages.
- You demonstrate how design and testing decisions interact to produce quality outcomes.
- You support your quality diagnosis using credible external evidence (e.g., peer-reviewed research, reputable white/green papers, benchmarking studies, standards/frameworks) and explain how it applies to your organisational SDLC context.
- To take this further, critically evaluate current practice against professional standards and emerging approaches (e.g., shift-left testing, architectural review practices) and synthesise an evidence-based quality diagnosis that produces original insight and motivates the project scope.

A - You made it!

- You critically evaluate current practice against professional standards and emerging approaches.
- You synthesise an evidence-based quality diagnosis that produces original insight into the organisation’s quality position and clearly motivates the project’s scope and direction.
- You synthesise evidence from multiple credible sources, compare viewpoints, and justify trade-offs to produce original insight into quality position and how it should shape the project’s scope and direction.

LO2 - Evaluate architectural characteristics, including coupling and design patterns, to justify improvements that increase system testability

For this learning outcome, ...

F - Failed for now

- You don't evaluate design or architectural characteristics. You don't identify recognisable design patterns or quality implications.

E - Almost at the passing grade

- You identify design patterns by name, but it's not enough to evidence their architectural role, quality implications, or relevance to the organisational context.

D - Congratulations, you passed!

- You describe design patterns and architectural characteristics present in the system.
- You outline quality or testability implications, but you don't justify recommendations against business context or assess proportionality.
- To strengthen this, explain the principles behind selected patterns and their architectural role, and recommend improvements with reference to business requirements and engineering principles.

C - Now we're climbing the ladder!

- You explain the principles behind selected design patterns and their architectural role.
- You identify quality and testability implications and recommend improvements with reference to business requirements and engineering principles.
- To move this on, analyse how design decisions (pattern selection, coupling, structural choices) interact to affect quality and testability, justify recommendations with context-specific evidence, and show proportionality (when simpler options are preferable).

B - Nearly at the top!

- You analyse how design decisions (pattern selection, coupling, structural choices) interact to affect software quality and testability.
- You justify recommendations with context-specific evidence and demonstrate proportionality in your decision-making.
- You support your architectural evaluation using credible external evidence (e.g., peer-reviewed research, reputable white/green papers, benchmarking studies, standards/frameworks) and explain how it strengthens your judgement about proportionality and testability improvements.
- To take this further, critically evaluate the design against organisational quality requirements and synthesise an architectural decision record showing sophisticated judgement (e.g., where patterns add unnecessary complexity, where trade-offs need explicit reasoning).

A - You made it!

- You critically evaluate the system's design against organisational quality requirements.
- You synthesise an architectural decision record that demonstrates sophisticated professional judgement, including explicit trade-off reasoning where competing design forces apply.
- You synthesise evidence from multiple credible sources, compare approaches, and justify trade-offs to demonstrate sophisticated architectural judgement in your decision record (e.g., when patterns add complexity vs improve testability).

LO3 - Implement design changes and deploy an automated testing strategy to evaluate quality across multiple stages of the development lifecycle

For this learning outcome, ...

F - Failed for now

- You don't implement design improvements or execute automated tests. You don't apply recognisable testing techniques or connect testing to the SDLC.

E - Almost at the passing grade

- You produce a limited set of tests, but it's not enough to evidence systematic technique selection or methodology awareness. The testing approach is not connected to documented design decisions.

D - Congratulations, you passed!

- You create and execute automated tests using at least one technique.
- You describe the testing approach, but you don't explicitly connect it to the design decisions made, evaluate effectiveness, or show refinement based on outcomes.
- To strengthen this, explain how the testing strategy responds to your documented design decisions, apply both black-box and white-box techniques, integrate with CI/CD tooling, and report results with coverage awareness.

C - Now we're climbing the ladder!

- You explain how the testing strategy responds to design decisions documented.
- You apply both black-box and white-box techniques, identify critical functionalities and edge cases, integrate testing with CI/CD tooling, and report results with coverage awareness.
- To move this on, analyse outcomes and refine the approach based on results, evaluate the effectiveness of techniques used, identify coverage limitations, and connect findings explicitly to the quality improvements you were aiming for.

B - Nearly at the top!

- You analyse testing outcomes and refine the approach based on results.
- You evaluate the effectiveness of techniques used, identify limitations in coverage, and connect findings explicitly to the quality improvements sought.
- You support your testing strategy evaluation using credible external evidence (e.g., peer-reviewed research, reputable white/green papers, benchmarking studies, standards/frameworks) and explain how it informed technique selection, refinement, and coverage decisions.
- To take this further, critically evaluate the testing strategy as a whole (what it proved, what it missed, what it means in context) and synthesise evidence-based conclusions that connect design decisions, testing outcomes, and quality impact in an original and professionally convincing way.

A - You made it!

- You critically evaluate the testing strategy as a whole and synthesise evidence-based conclusions that connect design decisions, testing outcomes, and quality impact in an original and professionally convincing way.
- You synthesise evidence from multiple credible sources, compare testing approaches, and justify trade-offs to connect design decisions, testing outcomes, and quality impact in a way that is convincing for your organisational context.

LO4 - Present technical design decisions and quality outcomes to both technical and non-technical stakeholders to demonstrate professional collaboration

For this learning outcome, ...

F - Failed for now

- You don't document stakeholder communication or professional standards. You show no awareness of collaborative practice or team working.

E - Almost at the passing grade.

- You provide basic documentation and state that communication took place, but it's not enough to evidence how decisions/outcomes were conveyed or how collaboration shaped the project.

D - Congratulations, you've passed!

- You describe communication with at least one stakeholder group and identify professional standards applied during the project.
- You don't explain how communication was adapted for different audiences or how collaboration influenced design or testing decisions.
- To strengthen this, explain how design decisions and quality outcomes were communicated to both technical and non-technical stakeholders and describe at least two collaborative practices (e.g., design critique) with their impact on the project.

C - Now we're climbing the ladder!

- You explain how design decisions and quality outcomes were communicated to both technical and non-technical stakeholders.
- You describe at least two collaborative practices with reference to their impact on the project and evidence of professional standards applied consistently.
- To move this on, analyse the effectiveness of communication and collaboration with specific examples of how feedback influenced design/testing decisions, and show sustained commitment to quality throughout the lifecycle.

B - Nearly at the top!

- You analyse the effectiveness of stakeholder communication and collaborative practice, with specific examples of how feedback influenced design or testing decisions.
- You demonstrate sustained commitment to quality throughout the project lifecycle and apply professional standards with clear awareness of their purpose.
- You support your communication and collaboration approach using credible external evidence (e.g., peer-reviewed research, reputable white/green papers, benchmarking studies, professional standards/frameworks) and explain how it informed audience adaptation and collaborative practices.
- To take this further, critically evaluate your communication and collaboration strategy as a whole, synthesising what worked, what you would do differently, and how your approach reflects and develops senior professional standards.

A - You made it!

- You critically evaluate the communication and collaboration strategy as a whole, synthesising insights into professional practice and how it shaped outcomes, including what you would do differently.
- You synthesise evidence from multiple credible sources and compare approaches to justify trade-offs in communication and collaboration strategy, showing how professional practice shaped outcomes and what you would do differently.

Glossary

Assessment brief: This is the assessment document that you get as part of every module. It contains some important information about the context of the assessment, a suggested structure as well as the assessment rubric.

Assessment Rubric: This is the piece of the assessment brief that houses the criteria against which you will be assessed. It is located at the end of the assessment brief document and is stored in a red, yellow and green matrix. Each of the learning outcomes are listed in this rubric together with what you have to do in order to achieve a certain grade.

LO / Learning Outcome: This refers to one of the four criteria that you will be assessed against. These are numbered and you will find them listed at the start of your assessment brief as well as in the assessment rubric. You will have 4 of these per assignment and they are equally weighted (each is worth 25% of the assignment)

KSB / Apprenticeship Standard: Your degree is aligned with the Digital and Technology Solutions Professional standard. This apprenticeship standard comes with a set of Knowledge, Skills and Behaviours that you must evidence in order to be awarded your degree apprenticeship. In each of your Level 5 and 6 assignments the learning outcomes are aligned with one or more of the KSBs. A D or above in a learning outcome tells you that you have sufficiently evidenced the associated KSBs. If you fail a learning outcome, you will need to resubmit the associated piece of work to ensure that you have sufficient evidence for all KSBs when it comes to your gateway.

Assessment Criteria: An individual cell in the assessment rubric expressing what is required for a particular grade.